Geniş başlıklı
Avrupa kesit profiller

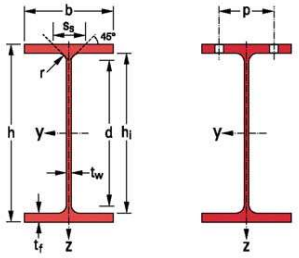
HE A, HE B et HE M 100 - 1000

Euronorm 53-62

HE AA 100 - 1000

HL 1000 / 1100

HEA
HEB
HEM



$$J = \frac{2bt^3 + d'w^3}{3}$$ (Galambos 1968)

$$C_w = \frac{(d')^2 b^3 t}{24}$$ (Galambos 1968, Picard and Beaulieu 1991)

	KESİT	ÖLÇÜLER					KESİT ALANI	TASARIM ÖLÇÜLERİ		KUVVETLİ EKSEN y-y					ZAYIF EKSEN z-z							SINIFLAMA ENV 1993 1-1						HİSTAR	yarım profillerde		
																							EĞİLME			BASINÇ				yg	J	Cw
		G kg/m	h mm	b mm	t_w mm	t_f mm	r mm	A cm^2	h_i mm	d mm	G kg/m	I_y cm^4	$W_{el.y}$ cm^3	$W_{pl.y}$ cm^3	i_y cm	A_{vz} cm^2	I_z cm^4	$W_{el.z}$ cm^3	$W_{pl.z}$ cm^3	i_z cm	s_s mm	I_t cm^4	$I_w x10^{-3}$ cm^6	235	355	460	235	355	460		mm	cm^4	cm^6
1	HE 100 AA	12.2	91	100	4.2	5.5	12	15.60	80	56	12.2	236.5	51.98	58.36	3.89	6.15	92.06	18.41	28.44	2.43	29.26	2.51	1.68	1	3	-	1	3	-		8.1	0.7	1.2
2	HE 100 A	16.7	96	100	5.0	8.0	12	21.24	80	56	16.7	349.2	72.76	83.01	4.06	7.56	133.8	26.76	41.14	2.51	35.06	5.24	2.58	1	1	-	1	1	-		8.9	1.9	3.6
3	HE 100 B	20.4	100	100	6.0	10.0	12	26.04	80	56	20.4	449.5	89.91	104.20	4.16	9.04	167.3	33.45	51.42	2.53	40.06	9.25	3.38	1	1	-	1	1	-		10.0	3.7	7.0
4	HE 100 M	41.8	120	106	12.0	20.0	12	53.24	80	56	41.8	1143.0	190.4	235.80	4.63	18.04	399.2	75.31	116.30	2.74	66.06	68.21	9.93	1	1	-	1	1	-		15.7	31.1	66.4
6	HE 120 AA	14.6	109	120	4.2	5.5	12	18.55	98	74	14.6	413.4	75.85	84.12	4.72	6.90	158.8	26.47	40.62	2.93	29.26	2.78	4.24	2	3	-	2	3	-		9.2	0.8	2.0
7	HE 120 A	19.9	114	120	5.0	8.0	12	25.34	98	74	19.9	606.2	106.3	119.5	4.89	8.46	230.9	38.48	58.85	3.02	35.06	5.99	6.47	1	1	-	1	1	-		9.8	2.3	6.2
8	HE 120 B	26.7	120	120	6.5	11.0	12	34.01	98	74	26.7	864.4	144.1	165.2	5.04	10.96	317.5	52.92	80.97	3.06	42.56	13.84	9.41	1	1	-	1	1	-		11.4	5.8	16.0
9	HE 120 M	52.1	140	126	12.5	21.0	12	66.41	98	74	52.1	2018.0	288.2	350.6	5.51	21.15	702.8	111.60	171.60	3.25	68.56	91.66	24.79	1	1	-	1	1	-		17.2	42.8	129.0
11	HE 140 AA	18.1	128	140	4.3	6.0	12	23.02	116	92	18.1	719.5	112.4	123.8	5.59	7.92	274.8	39.26	59.93	3.45	30.36	3.54	10.21	3	3	-	3	3	-		10.2	1.2	4.1
12	HE 140 A	24.7	133	140	5.5	8.5	12	31.42	116	92	24.7	1033.0	155.4	173.5	5.73	10.12	389.3	55.62	84.85	3.52	36.56	8.13	15.06	1	2	-	1	2	-		11.3	3.2	11.7
13	HE 140 B	33.7	140	140	7.0	12.0	12	42.96	116	92	33.7	1509.0	215.6	245.4	5.93	13.08	549.7	78.52	119.80	3.58	45.06	20.06	22.48	1	1	-	1	1	-		12.9	8.8	33.0
14	HE 140 M	63.2	160	146	13.0	22.0	12	80.56	116	92	63.2	3291.0	411.4	493.8	6.39	24.46	1144.0	156.80	240.50	3.77	71.06	120.00	54.33	1	1	-	1	1	-		18.7	56.9	230.5
16	HE 160 AA	23.8	148	160	4.5	7.0	15	30.36	134	104	23.8	1283.0	173.4	190.4	6.50	10.38	478.7	59.84	91.36	3.97	36.07	6.33	23.75	3	3	-	3	3	-		11.3	2.0	9.8
17	HE 160 A	30.4	152	160	6.0	9.0	15	38.77	134	104	30.4	1673.0	220.1	245.1	6.57	13.21	615.6	76.95	117.6	3.98	41.57	12.19	31.41	1	2	-	1	2	-		12.8	4.4	20.8
18	HE 160 B	42.6	160	160	8.0	13.0	15	54.25	134	104	42.6	2492.0	311.5	354.0	6.78	17.59	889.2	111.2	170.0	4.05	51.57	31.24	47.94	1	1	-	1	1	-		14.7	13.0	62.6
19	HE 160 M	76.2	180	166	14.0	23.0	15	97.05	134	104	76.2	5098.0	566.5	674.6	7.25	30.81	1759.0	211.9	325.5	4.26	77.57	162.40	108.10	1	1	-	1	1	-		20.5	74.5	387.1
21	HE 180 AA	28.7	167	180	5.0	7.5	15	36.53	152	122	28.7	1967.0	235.6	258.2	7.34	12.16	730.0	81.1	123.6	4.47	37.57	8.33	46.36	3	3	-	3	3	-		12.8	2.9	17.1
22	HE 180 A	35.5	171	180	6.0	9.5	15	45.25	152	122	35.5	2510.0	293.6	324.9	7.45	14.47	924.6	102.7	156.5	4.52	42.57	14.80	60.21	1	3	-	1	3	-		13.7	5.7	34.8
23	HE 180 B	51.2	180	180	8.5	14.0	15	65.25	152	122	51.2	3831.0	425.7	481.4	7.66	20.24	1363.0	151.4	231.0	4.57	54.07	42.16	93.75	1	1	-	1	1	-		16.2	18.2	111.3
24	HE 180 M	88.9	200	186	14.5	24.0	15	113.30	152	122	88.9	7483.0	748.3	883.4	8.13	34.65	2580	277.4	425.2	4.77	80.07	203.30	199.30	1	1	-	1	1	-		22.0	94.7	618.5
26	HE 200 AA	34.6	186	200	5.5	8.0	18	44.13	170	134	34.6	2944.0	316.6	347.1	8.17	15.45	1068.0	106.8	163.2	4.92	42.59	12.69	84.49	3	4	-	3	4	-		14.3	3.9	28.5
27	HE 200 A	42.3	190	200	6.5	10.0	18	53.83	170	134	42.3	3692.0	388.6	429.5	8.28	18.08	1336.0	133.6	203.8	4.98	47.59	20.98	108.0	1	3	-	1	3	-		15.2	7.5	55.6
28	HE 200 B	61.3	200	200	9.0	15.0	18	78.08	170	134	61.3	5696.0	569.6	642.5	8.54	24.83	2003.0	200.3	305.8	5.07	60.09	59.28	171.1	1	1	-	1	1	-		17.7	24.7	187.7
29	HE 200 M	103.0	220	206	15.0	25.0	18	131.30	170	134	103.0	10640.0	967.4	1135.0	9.00	41.03	3651.0	354.5	543.2	5.27	86.09	259.40	346.3	1	1	-	1	1	-		23.6	118.3	949.5
31	HE 220 AA	40.4	205	220	6.0	8.5	18	51.46	188	152	40.4	4170.0	406.9	445.5	9.00	17.63	1510.0	137.3	209.3	5.42	44.09	15.93	145.6	3	4	-	3	4	-		15.9	5.2	45.5
32	HE 220 A	50.5	210	220	7.0	11.0	18	64.34	188	152	50.5	5410.0	515.2	568.5	9.17	20.67	1955	177.7	270.6	5.51	50.09	28.46	193.3	1	3	-	1	3	-		16.6	10.9	98.5
33	HE 220 B	71.5	220	220	9.5	16.0	18	91.04	188	152	71.5	8091.0	735.5	827.0	9.43	27.92	2843.0	258.5	393.9	5.59	62.59	76.57	295.4	1	1	-	1	1	-		19.2	33.0	303.1
34	HE 220 M	117.0	240	226	15.5	26.0	18	149.40	188	152	117.0	14600.0	1217.0	1419.0	9.89	45.31	5012.0	443.5	678.6	5.79	88.59	315.30	572.7	1	1	-	1	1	-		25.0	145.7	1410.0
36	HE 240 AA	47.4	224	240	6.5	9.0	21	60.38	206	164	47.4	5835.0	521.0	570.6	9.83	21.54	2077	173.1	264.4	5.87	49.10	22.98	239.6	3	4	-	3	4	-		17.5	6.8	70.1
37	HE 240 A	60.3	230	240	7.5	12.0	21	76.84	206	164	60.3	7763.0	675.1	744.6	10.05	25.18	2769	230.7	351.7	6.00	56.10	41.55	328.5	1	3	-	1	3	-		18.1	15.4	166.0
38	HE 240 B	83.2	240	240	10.0	17.0	21	106.00	206	164	83.2	11260.0	938.3	1053.0	10.31	33.23	3923	326.9	498.4	6.08	68.60	102.70	486.9	1	1	-	1	1	-		20.7	43.0	472.0
39	HE 240 M	157.0	270	248	18.0	32.0	21	199.60	206	164	157.0	24290.0	1799.0	2117.0	11.03	60.07	8153	657.5	1006.0	6.39	106.60	627.90	1152.0	1	1	-	1	1	-		28.9	294.0	3472.8

41	HE 260 AA	54.1	244	260	6.5	9.5	24	68.97	225	177	54.1	7981.0	654.1	714.5	10.76	24.75	2788	214.5	327.7	6.36	53.62	30.31	382.6	3	4	-	3	4	-		18.4	8.5	104.7
42	HE 260 A	68.2	250	260	7.5	12.5	24	86.82	225	177	68.2	10450.0	836.4	919.8	10.97	28.76	3668	282.1	430.2	6.50	60.62	52.37	516.4	2	3	3	2	3	3	HI	19.1	18.6	238.5
43	HE 260 B	93.0	260	260	10.0	17.5	24	118.40	225	177	93.0	14920.0	1148.0	1283.0	11.22	37.59	5135	395.0	602.2	6.58	73.12	123.80	753.7	1	1	2	1	1	2	HI	21.6	50.5	654.5
44	HE 260 M	172	290	268	18.0	32.5	24	219.60	225	177	172.0	31310.0	2159.0	2524.0	11.94	66.89	10450	779.7	1192.0	6.90	111.10	719.00	1728.0	1	1	1	1	1	1	HI	30.1	331.7	4590.8
46	HE 280 AA	61.2	264	280	7.0	10.0	24	78.02	244	196	61.2	10560.0	799.8	873.1	11.63	27.52	3664	261.7	399.4	6.85	55.12	36.22	590.1	3	4	-	3	4	-		20.1	10.8	152.6
47	HE 280 A	76.4	270	280	8.0	13.0	24	97.26	244	196	76.4	13670.0	1013.0	1112	11.86	31.74	4763	340.2	518.1	7.00	62.12	62.10	785.4	2	3	4	2	3	4	HI	20.7	22.7	335.1
48	HE 280 B	103	280	280	10.5	18.0	24	131.40	244	196	103.0	19270.0	1376.0	1534	12.11	41.09	6595	471.0	717.6	7.09	74.62	143.70	1130.0	1	1	2	1	1	2	HI	23.3	59.5	889.5
49	HE 280 M	189.0	310	288	18.5	33.0	24	240.20	244	196	189.0	39550.0	2551.0	2966.0	12.83	72.03	13160	914.1	1397.0	7.40	112.60	807.30	2520.0	1	1	1	1	1	1	HI	31.5	374.2	5964.0
51	HE 300 AA	69.8	283	300	7.5	10.5	27	88.91	262	208	69.8	13800.0	976	1065	12.46	32.37	4734	315.6	482.3	7.30	60.13	49.35	877.2	3	4	-	3	4	-		21.7	13.5	217.2
52	HE 300 A	88	290	300	8.5	14.0	27	112.50	262	208	88.3	18260.0	1260	1383	12.74	37.28	6310	420.6	641.2	7.49	68.13	85.17	1200	2	3	3	2	3	3	HI	22.1	30.3	514.7
53	HE 300 B	117.0	300	300	11.0	19.0	27	149.10	262	208	117.0	25170.0	1678.0	1869.0	12.99	47.43	8563	570.9	870.1	7.58	80.63	185.00	1688	1	1	3	1	1	3	HI	24.6	74.8	1286.6
54	HE 300 M	238.0	340	310	21.0	39.0	27	303.10	262	208	238.0	59200.0	3482	4078	13.98	90.53	19400	1252.0	1913.0	8.00	130.60	1408.00	4386	1	1	1	1	1	1	HI	35.5	659.4	12275.9
56	HE 320 AA	74	301	300	8.0	11.0	27	94.58	279	225	74.2	16450.0	1093	1196	13.19	35.40	4959	330.6	505.7	7.24	61.63	55.87	1041	3	4	-	3	4	-		24.0	15.8	249.8
57	HE 320 A	97.6	310	300	9.0	15.5	27	124.40	279	225	97.6	22930.0	1479.0	1628	13.58	41.13	6985	465.7	709.7	7.49	71.63	108.00	1512	1	3	3	1	3	3	HI	24.1	40.8	698.5
58	HE 320 B	127.0	320	300	11.5	20.5	27	161.30	279	225	127.0	30820.0	1926	2149	13.82	51.77	9239	615.9	939.1	7.57	84.13	225.10	2069	1	1	2	1	1	2	HI	26.8	93.7	1616.0
59	HE 320 M	245	359	309	21.0	40.0	27	312.00	279	225	245.0	68130.0	3796	4435	14.78	94.85	19710	1276.0	1951.0	7.95	132.60	1501.00	5004	1	1	1	1	1	1	HI	37.4	708.4	13116.8
61	HE 340 AA	78.9	320	300	8.5	11.5	27	100.50	297	243	78.9	19550.0	1222	1341	13.95	38.69	5185	345.6	529.3	7.18	63.13	63.07	1231	3	4	-	3	4	-		26.6	18.4	285.4
62	HE 340 A	105.0	330	300	9.5	16.5	27	133.50	297	243	105.0	27690.0	1678	1850	14.40	44.95	7436	495.7	755.9	7.46	74.13	127.20	1824	1	3	3	1	3	3	HI	26.4	49.4	842.6
63	HE 340 B	134	340	300	12.0	21.5	27	170.90	297	243	134.0	36660.0	2156	2408	14.65	56.09	9690	646.0	985.7	7.53	86.63	257.20	2454	1	1	1	1	1	1	HI	29.1	108.6	1864.2
64	HE 340 M	248	377	309	21.0	40.0	27	315.80	297	243	248.0	76370.0	4052	4718	15.55	98.63	19710	1276.0	1953.0	7.90	132.60	1506.00	5584	1	1	1	1	1	1	HI	39.1	711.2	13117.1
66	HE 360 AA	84	339	300	9.0	12.0	27	106.60	315	261	83.7	23040.0	1359	1495	14.70	42.17	5410	360.7	553.0	7.12	64.63	70.99	1444	3	4	-	3	4	-		29.3	21.3	324.3
67	HE 360 A	112	350	300	10.0	17.5	27	142.80	315	261	112.0	33090.0	1891	2088	15.22	48.96	7887	525.8	802.3	7.43	76.63	148.80	2177	1	2	3	1	2	3	HI	28.8	59.1	1005.3
68	HE 360 B	142	360	300	12.5	22.5	27	180.60	315	261	142.0	43190.0	2400	2683	15.46	60.60	10140	676.1	1032.0	7.49	89.13	292.50	2883	1	1	1	1	1	1	HI	31.4	124.9	2136.7
69	HE 360 M	250.0	395	308	21.0	40.0	27	318.80	315	261	250.0	84870.0	4297	4989	16.32	102.40	19520	1268.0	1942.0	7.83	132.60	1507.00	6137	1	1	1	1	1	1	HI	41.0	711.9	12990.4
71	HE 400 AA	92	378	300	9.5	13.0	27	117.70	352	298	92.4	31250.0	1654	1824	16.30	47.95	5861	390.8	599.7	7.06	67.13	84.69	1948	3	3	-	3	3	-		34.0	27.2	412.4
72	HE 400 A	125	390	300	11.0	19.0	27	159.00	352	298	125.0	45070.0	2311	2562	16.84	57.33	8564	570.9	872.9	7.34	80.63	189.00	2942	1	1	3	1	2	3	HI	33.9	76.8	1286.7
73	HE 400 B	155.0	400	300	13.5	24.0	27	197.80	352	298	155.0	57680.0	2884	3232	17.08	69.98	10820	721.3	1104.0	7.40	93.13	355.70	3817	1	1	1	1	1	1	HI	36.6	153.7	2593.3
74	HE 400 M	256	432	307	21.0	40.0	27	325.80	352	298	256.0	104100.0	4820	5571	17.88	110.20	19340	1260.0	1934.0	7.70	132.60	1515.00	7410	1	1	1	1	1	1	HI	45.0	715.4	12864.8
76	HE 450 AA	100	425	300	10.0	13.5	27	127.10	398	344	99.7	41890.0	1971	2183	18.16	54.70	6088	405.8	624.4	6.92	68.63	95.61	2572	3	3	-	3	4	-		40.7	31.5	461.9
77	HE 450 A	140	440	300	11.5	21.0	27	178.00	398	344	140.0	63720.0	2896	3216	18.92	65.78	9465	631.0	965.5	7.29	85.13	243.80	4148	1	1	1	1	2	3	HI	39.3	103.2	1737.3
78	HE 450 B	171	450	300	14.0	26.0	27	218.00	398	344	171.0	79890.0	3551	3982	19.14	79.66	11720	781.4	1198.0	7.33	97.63	440.50	5258	1	1	1	1	1	2	HI	42.3	195.2	3297.1
79	HE 450 M	263	478	307	21.0	40.0	27	335.40	398	344	263.0	131500.0	5501	6331	19.80	119.80	19340	1260.0	1939.0	7.59	132.60	1529.00	9251	1	1	1	1	1	1	HI	50.2	722.5	12865.4
81	HE 500 AA	107	472	300	10.5	14.0	27	136.90	444	390	107.0	54640.0	2315	2576	19.98	61.91	6314	420.9	649.3	6.79	70.13	107.70	3304	2	3	-	2	4	-		47.8	36.3	515.2
82	HE 500 A	155	490	300	12.0	23.0	27	197.50	444	390	155.0	86970.0	3550	3949	20.98	74.72	10370	691.1	1059.0	7.24	89.63	309.30	5643	1	1	1	1	3	4	HI	45.1	135.1	2282.4
83	HE 500 B	187	500	300	14.5	28.0	27	238.60	444	390	187.0	107200.0	4287	4815	21.19	89.82	12620	841.6	1292.0	7.27	102.10	538.40	7018	1	1	1	1	2	2	HI	48.2	243.5	4118.0
84	HE 500 M	270	524	306	21.0	40.0	27	344.30	444	390	270.0	161900.0	6180	7094	21.69	129.50	19150	1252.0	1932.0	7.46	132.60	1539.00	11190	1	1	1	1	1	1	HI	56.0	727.5	12740.7
86	HE 550 AA	120	522	300	11.5	15.0	27	152.80	492	438	120.0	72870.0	2792	3128	21.84	72.66	6767	451.1	698.6	6.65	73.13	133.70	4338	1	3	-	3	4	-		56.3	46.6	633.9
87	HE 550 A	166	540	300	12.5	24.0	27	211.80	492	438	166.0	111900.0	4146	4622	22.99	83.72	10820	721.3	1107.0	7.15	92.13	351.50	7189	1	1	1	2	4	4	HI	51.8	155.0	2593.4
88	HE 550 B	199	550	300	15.0	29.0	27	254.10	492	438	199.0	136700.0	4971	5591	23.20	100.10	13080	871.8	1341.0	7.17	104.60	600.30	8856	1	1	1	1	2	3	HI	55.0	273.2	4575.4
89	HE 550 M	278	572	306	21.0	40.0	27	354.40	492	438	278.0	198000.0	6923	7933	23.64	139.60	19160	1252.0	1937.0	7.35	132.60	1554.00	13520	1	1	1	1	1	1	HI	62.2	734.9	12741.3
91	HE 600 AA	129	571	300	12.0	15.5	27	164.10	540	486	129.0	91900.0	3218	3623	23.66	81.29	6993	466.2	724.5	6.53	74.63	149.80	5381	1	3	-	3	4	-		64.7	53.2	699.6
92	HE 600 A	178	590	300	13.0	25.0	27	226.50	540	486	178.0	141200.0	4787	5350	24.97	93.21	11270	751.4	1156.0	7.05	94.63	397.80	8978	1	1	1	2	4	4	HI	58.8	176.9	2931.4
93	HE 600 B	212	600	300	15.5	30.0	27	270.00	540	486	212.0	171000.0	5701	6425	25.17	110.80	13530	902.0	1391.0	7.08	107.10	667.20	10970	1	1	1	1	3	4	HI	62.0	305.4	5065.4
94	HE 600 M	285	620	305	21.0	40.0	27	363.70	540	486	285.0	237400.0	7660	8772	25.55	149.70	18980	1244.0	1930.0	7.22	132.60	1564.00	15910	1	1	1	1	1	1	HI	68.8	740.2	12617.5
95	HE 600 x 337	337	632	310	25.5	46.0	27	429.20	540	486	337.0	283200.0	8961	10380	25.69	180.50	22940	1480.0	2310.0	7.31	149.10	2451.00	19610	1	1	1	1	1	1	HI	74.2	1167.8	20150.6
96	HE 600 x 399	399	648	315	30.0	54.0	27	508.50	540	486	399.0	344600.0	10640	12460	26.03	213.60	28280	1796.0	2814.0	7.46	169.60	3966.00	24810	1	1	1	1	1	1	HI	79.0	1920.7	34200.6
98	HE 650 AA	138	620	300	12.5	16.0	27	175.80	588	534	138.0	113900.0	3676	4160	25.46	90.40	7221	481.4	750.7	6.41	76.13	167.50	6567	1	3	-	4	4	-		73.4	60.6	769.6
99	HE 650 A	190	640	300	13.5	26.0	27	241.60	588	534	190.0	175200.0	5474	6136	26.93	103.20	11720	781.6	1205.0	6.97	97.13	448.30	11030	1	1	1	3	4	4	HI	66.0	200.9	3297.6
100	HE 650 B	225	650	300	16.0	31.0	27	286.30	588	534	225.0	210600.0	6480	7320	27.12	122.00	13980	932.3	1441.0	6.99	109.60	739.20	13360	1	1	1	2	3	4	HI	69.3	340.2	5589.3

101	HE 650 M	293	668	305	21.0	40.0	27	373.70	588	534	293.0	281700.0	8433	9657	27.45	159.70	18980	1245.0	1936.0	7.13	132.60	1579.00	18650	1	1	1	1	1	2	HI	75.6	747.6	12618.1
102	HE 650 x 343	343	680	309	25.0	46.0	27	437.50	588	534	343.0	333700.0	9815	11350	27.62	189.60	22720	1470.0	2300.0	7.21	148.60	2442.00	22730	1	1	1	1	1	1	HI	80.6	1167.7	19956.6
103	HE 650 x 407	407	696	314	29.5	54.0	27	518.80	588	534	407.0	405400.0	11650	13620	27.95	224.80	28020	1785.0	2803.0	7.35	169.10	3958.00	28710	1	1	1	1	1	1	HI	85.5	1922.8	33876.7
105	HE 700 AA	150	670	300	13.0	17.0	27	190.90	636	582	150.0	142700.0	4260	4840	27.34	100.30	7673	511.5	799.7	6.34	78.63	195.20	8155	1	2	-	4	4	-		81.5	73.0	923.2
106	HE 700 A	204	690	300	14.5	27.0	27	260.50	636	582	204.0	215300.0	6241	7032	28.75	117.00	12180	811.9	1257.0	6.84	100.10	513.90	13350	1	1	1	3	4	4	HI	75.1	230.5	3693.4
107	HE 700 B	241	700	300	17.0	32.0	27	306.40	636	582	241.0	256900.0	7340	8327	28.96	137.10	14440	962.7	1495.0	6.87	112.60	830.90	16060	1	1	1	2	4	4	HI	78.2	382.4	6148.6
108	HE 700 M	301	716	304	21.0	40.0	27	383.00	636	582	301.0	329300.0	9198	10540	29.32	169.80	18800	1237.0	1929.0	7.01	132.60	1589.00	21400	1	1	1	1	2	3	HI	82.8	752.9	12495.1
109	HE 700 x 352	352	728	308	25.0	46.0	27	448.60	636	582	352.0	389700.0	10710	12390	29.47	201.60	22510	1461.0	2293.0	7.08	148.60	2461.00	26050	1	1	1	1	1	1	HI	87.8	1176.9	19764.6
110	HE 700 x 418	418	744	313	29.5	54.0	27	531.90	636	582	418.0	472500.0	12700	14840	29.80	239.00	27760	1774.0	2797.0	7.22	169.10	3989.00	32850	1	1	1	1	1	1	HI	93.0	1938.1	33556.0
112	HE 800 AA	172	770	300	14.0	18.0	30	218.50	734	674	172.0	208900.0	5426	6225	30.92	123.80	8134	542.2	856.6	6.10	85.15	256.80	11450	1	2	-	4	4	-		100.1	92.7	1096.4
113	HE 800 A	224	790	300	15.0	28.0	30	285.80	734	674	224.0	303400.0	7682	8699	32.58	138.80	12640	842.6	1312.0	6.65	106.10	596.90	18290	1	1	1	4	4	4	HI	90.6	262.4	4119.6
114	HE 800 B	262	800	300	17.5	33.0	30	334.20	734	674	262.0	359100.0	8977	10230	32.78	161.80	14900	993.6	1553.0	6.68	118.60	946.00	21840	1	1	1	3	4	4	HI	93.9	427.9	6743.9
115	HE 800 M	317	814	303	21.0	40.0	30	404.30	734	674	317.0	442600.0	10870	12490	33.09	194.30	18630	1230.0	1930.0	6.79	136.10	1646.00	27780	1	1	1	1	3	4	HI	98.1	765.9	12373.6
116	HE 800 x 373	373	826	308	25.0	46.0	30	474.60	734	674	373.0	523900.0	12690	14700	33.23	230.30	22530	1463.0	2311.0	6.89	152.10	2554.00	34070	1	1	1	1	2	2	HI	103.3	1202.4	19766.7
117	HE 800 x 444	444	842	313	30.0	54.0	30	566.00	734	674	444.0	634500.0	15070	17640	33.48	276.50	27800	1776.0	2827.0	7.01	173.10	4180.00	42840	1	1	1	1	1	1	HI	109.3	1997.5	33561.0
119	HE 900 AA	198	870	300	15.0	20.0	30	252.20	830	770	198.0	301100.0	6923	7999	34.55	147.20	9041	602.8	957.7	5.99	90.15	334.90	16260	1	1	-	4	4	-		117.8	127.8	1504.0
120	HE 900 A	252	890	300	16.0	30.0	30	320.50	830	770	252.0	422100.0	9485	10810	36.29	163.30	13550	903.2	1414.0	6.50	111.10	736.80	24960	1	1	1	4	4	4	HI	107.7	328.7	5067.4
121	HE 900 B	291	900	300	18.5	35.0	30	371.30	830	770	291.0	494100.0	10980	12580	36.48	188.80	15820	1054.0	1658.0	6.53	123.60	1137.00	29460	1	1	1	3	4	4	HI	111.2	520.0	8046.7
122	HE 900 M	333	910	302	21.0	40.0	30	423.60	830	770	333.0	570400.0	12540	14440	36.70	214.40	18450	1222.0	1929.0	6.60	136.10	1671.00	34750	1	1	1	2	4	4	HI	114.1	778.6	12252.8
123	HE 900 x 391	391	922	307	25.0	46.0	30	497.70	830	770	391.0	674300.0	14630	16990	36.81	254.30	22320	1454.0	2312.0	6.70	152.10	2597.00	42560	1	1	1	1	3	4	HI	119.6	1224.2	19577.1
124	HE 900 x 466	466	938	312	30.0	54.0	30	593.70	830	770	466.0	814900.0	17380	20380	37.05	305.30	27560	1767.0	2832.0	6.81	173.10	4256.00	53400	1	1	1	1	1	2	HI	125.7	2035.4	33244.2
126	HE 1000 AA	222	970	300	16.0	21.0	30	282.20	928	868	222.0	406500.0	8380	9777	37.95	172.20	9501	633.4	1016.0	5.80	93.15	403.40	21280	1	1	-	4	4	-		138.5	157.4	1741.8
127	HE 1000 A	272	990	300	16.5	31.0	30	346.80	928	868	272.0	553800.0	11190	12820	39.96	184.60	14000	933.6	1470.0	6.35	113.60	822.40	32070	1	1	2	4	4	4	HI	125.3	369.7	5591.8
128	HE 1000 B	314	1000	300	19.0	36.0	30	400.00	928	868	314.0	644700.0	12890	14860	40.15	212.50	16280	1085.0	1716.0	6.38	126.10	1254.00	37640	1	1	1	4	4	4	HI	128.5	576.8	8757.2
129	HE 1000 M	349	1008	302	21.0	40.0	30	444.20	928	868	349.0	722300.0	14330	16570	40.32	235.00	18460	1222.0	1940.0	6.45	136.10	1701.00	43020	1	1	1	3	4	4	HI	131.0	793.7	12254.1
130	HE 1000 x 393	393	1016	303	24.4	43.9	30	500.20	928	868	393.0	807700.0	15900	18540	40.18	271.30	20500	1353.0	2168.0	6.40	147.30	2332.00	48080	1	1	1	2	4	4	HI	137.3	1089.9	16363.6
131	HE 1000 x 409	409	1020	306	25.0	46.0	30	521.20	928	868	409.0	850800.0	16680	19450	40.40	278.80	22120	1446.0	2313.0	6.51	152.10	2642.00	52100	1	1	1	2	4	4	HI	136.8	1246.5	19388.7
132	HE 1000 x 488	488	1036	311	30.0	54.0	30	622.00	928	868	488.0	1027000.0	19820	23300	40.62	334.70	27320	1757.0	2837.0	6.63	173.10	4334.00	65270	1	1	1	1	2	3	HI	143.4	2074.3	32929.6
133	HE 1000 x 579	579	1056	316	35.0	64.0	30	737.00	928	868	579.0	1246000.0	23590	27950	41.11	393.30	34040	2154.0	3498.0	6.80	198.10	7102.00	82800	1	1	1	1	1	2	HI	148.8	3470.1	57502.3
135	HL 920 x 342	342	912	418	19.3	32.0	24	436.10	848	800	342.0	624900.0	13700	15450	37.85	190.10	39010	1867.0	2882.0	9.46	111.40	1193.00	75410	1	1	1	3	4	4	HI	101.7	562.0	16628.2
136	HL 920 x 365	365	916	419	20.3	34.3	24	464.40	847.4	799.4	365.0	670500.0	14640	16520	38.00	200.40	42120	2011.0	3106.0	9.52	117.00	1446.00	81730	1	1	1	3	4	4	HI	102.3	686.5	20624.2
137	HL 920 x 387	387	921	420	21.3	36.6	24	493.00	847.8	799.8	387.0	718300.0	15600	17630	38.17	210.90	45280	2156.0	3332.0	9.58	122.60	1734.00	88370	1	1	1	2	4	4	HI	102.9	828.8	25236.7
138	HL 920 x 417	417	928	422	22.5	39.9	24	532.50	848.2	800.2	417.0	787600.0	16970	19210	38.46	223.90	50070	2373.0	3668.0	9.70	130.40	2200.00	98540	1	1	1	2	4	4	HI	103.2	1062.1	33164.8
139	HL 920 x 446	446	933	423	24.0	42.7	24	569.60	847.6	799.6	446.0	846800.0	18150	20600	38.56	239.10	53980	2552.0	3951.0	9.73	137.50	2685.00	106740	1	1	1	2	3	4	HI	104.8	1302.9	40937.7
140	HL 920 x 488	488	942	422	25.9	47.0	24	621.30	848	800	488.0	935390.0	19860	22615	38.80	259.30	59010	2797.0	4336.0	9.75	148.00	3514.00	117890	1	1	1	1	2	4	HI	107.0	1719.6	54205.3
141	HL 920 x 534	534	950	425	28.4	51.1	24	680.10	847.8	799.8	534.0	1031000.0	21710	24830	38.94	284.80	65560	3085.0	4796.0	9.82	158.70	4542.00	132070	1	1	1	1	2	3	HI	109.9	2233.5	71160.9
142	HL 920 x 585	585	960	427	31.0	55.9	24	745.30	848.2	800.2	585.0	1143090.0	23814	27363	39.16	312.00	72770	3408.0	5310.0	9.88	170.90	5932.00	148220	1	1	1	1	1	2	HI	112.9	2935.1	94477.5
143	HL 920 x 653	653	972	431	34.5	62.0	24	831.90	848	800	653.0	1292000.0	26590	30730	39.41	348.70	83050	3854.0	6022.0	9.99	186.60	8124.00	171280	1	1	1	1	1	1	HI	116.6	4046.8	132560.6
144	HL 920 x 784	784	996	437	40.9	73.9	24	997.70	848.2	800.2	784.0	1593000.0	31980	37340	39.95	417.60	103300	4728.0	7424.0	10.18	216.80	13730.00	218490	1	1	-	1	1	-		123.7	6930.3	233979.5
145	HL 920 x 967	967	1028	446	50.0	89.9	24	1231.00	848.2	800.2	967.0	2033000.0	39540	46810	40.64	517.10	133900	6003.0	9486.0	10.43	257.90	24930.00	292450	1	1	-	1	1	-		133.7	12756.1	447794.9
147	HL 1000 x 296	296	982	400	16.5	27.0	30	376.80	928	868	296.0	618700.0	12600	14220	40.52	181.50	28850	1443.0	2235.0	8.75	105.60	756.90	65670	1	1	2	4	4	4	HI	113.6	333.9	8754.0
148	HL 1000 A	321	990	400	16.5	31.0	30	408.80	928	868	321.0	696400.0	14070	15800	41.27	184.60	33120	1656.0	2555.0	9.00	113.60	1021.00	76030	1	1	2	4	4	4	HI	108.5	469.0	13246.4
149	HL 1000 B	371	1000	400	19.0	36.0	30	472.00	928	868	371.0	812100.0	16240	18330	41.48	212.50	38480	1924.0	2976.0	9.03	126.10	1565.00	89210	1	1	1	4	4	4	HI	111.7	732.3	20745.2
150	HL 1000 M	412	1008	402	21.0	40.0	30	524.20	928	868	412.0	909800.0	18050	20440	41.66	235.00	43410	2160.0	3348.0	9.10	136.10	2128.00	101460	1	1	1	3	4	4	HI	114.1	1007.0	28885.7
151	HL 1000 x 477	477	1018	404	25.5	45.0	30	608.00	928	868	477.0	1047000.0	20570	23530	41.50	282.80	49610	2456.0	3838.0	9.03	150.60	3159.00	117050	1	1	1	2	3	4	HI	122.0	1496.0	41749.6
152	HL 1000 x 554	554	1032	408	29.5	52.0	30	705.80	928	868	554.0	1232000.0	23880	27500	41.79	328.00	59100	2897.0	4547.0	9.15	168.60	4860.00	141330	1	1	1	1	2	3	HI	126.4	2331.6	66352.4
153	HL 1000 x 642	642	1048	412	34.0	60.0	30	817.60	928	868	642.0	1451000.0	27680	32100	42.12	379.60	70280	3412.0	5379.0	9.27	189.10	7440.00	170670	1	1	1	1	1	2	HI	131.4	3613.6	104955.7
154	HL 1000 x 748	748	1068	417	39.0	70.0	30	953.40	928	868	748.0	1732000.0	32430	37880	42.62	438.90	85111	4082.0	6459.0	9.45	214.10	11670.00	210650	1	1	1	1	1	1	HI	136.7	5754.4	172801.1
155	HL 1000 x 883	883	1092	424	45.5	82.0	30	1125.30	928	868	883.0	2096000.0	38390	45260	43.16	516.50	105000	4952.0	7874.0	9.66	244.60	18750.00	265670	1	1	-	1	1	-		143.8	9378.3	291993.5

157	HL 1100 A	343	1090	400	18.0	31.0	20	436.50	1028	988	343.0	867400.0	15920	18060	44.58	206.50	33120	1656.0	2568.0	8.71	103.40	1037.00	92710	1	1	2	4	4	4	HI	131.3	500.1	13249.0
158	HL 1100 B	390	1100	400	20.0	36.0	20	497.00	1028	988	390.0	1005000.0	18280	20780	44.98	230.60	38480	1924.0	2988.0	8.80	115.40	1564.00	108680	1	1	1	4	4	4	HI	131.9	763.9	20747.8
159	HL 1100 M	433	1108	402	22.0	40.0	20	551.20	1028	988	433.0	1126000.0	20320	23160	45.19	254.40	43410	2160.0	3362.0	8.87	125.40	2130.00	123500	1	1	1	4	4	4	HI	133.8	1047.1	28889.0
160	HL 1100 R	499	1118	405	26.0	45.0	20	635.20	1028	988	499.0	1294000.0	23150	26600	45.14	300.40	49980	2468.0	3870.0	8.87	139.40	3135.00	143410	1	1	1	2	4	4	HI	140.2	1544.5	42064.0